

Judgment Aggregation: Prediction Market or Majority Vote?

Korbinian Friedl

Table of contents

1	Knowing the Future	2
2	Example: Horse Bad?	3
2.1	Opinions Are Divided	4
3	In this Talk	5
3.1	In this Talk	6
4	Outline	6
5	What is the CJT and What Are Prediction Markets	6
5.1	General Setting	6
5.2	Assumptions Required for the CJT	6
5.3	Theorem 1 (Condorcet Jury Theorem)	7
5.4	Prediction Markets	7
5.5	Prediction Markets as Judgment Aggregators	7
5.6	Example	8
6	Modelling Prediction Markets as Exchange Economies	8
6.1	Definition 1 (Exchange Economy)	8
6.2	Definition 2 (Consumption Bundle)	8
6.3	Definition 3 (Allocation in an Exchange Economy)	9
6.4	Definition 4 (Feasibility of an Allocation)	9
6.5	Definition 5 (Walrasian Equilibrium)	9
7	Jury Theorems for Prediction Markets	9
7.1	Assumptions Required	9
7.2	Further Modelling Assumptions	9

8	Jury Theorems for Prediction Markets	10
8.1	Theorem 2 (PMJT for Agents with Binary Beliefs)	10
8.2	Theorem 3 (PMJT for Agents with Degrees of Belief)	10
8.3	Can We Recover the Condorcet Properties in this Case?	10
8.4	Corollary 4 (Condorcet-Like Property for PM, Degrees of Belief)	10
9	The Perspective of the “No Trade Theorem”	11
10	Conclusion	11
	References	11

1 Knowing the Future

- It sure would be nice to know the future some times.
- I mean not just because it would be interesting, but also because it could be really helpful for taking appropriate precautions or making the necessary preparations.

2 Example: Horse Bad?



Figure 1: (Tiepolo c. 1760)

2.1 Opinions Are Divided





3 In this Talk

- Will present theorems to the effect that (if modelled as exchange economies), prediction markets will exhibit the same advantages which the CJT certifies for majority voting
- In this sense, they are no less (and perhaps more) effective at tapping into individuals' rationality and expertise for the purpose of getting accurate predictions

3.1 In this Talk

- I will end, however, by briefly discussing another result, which suggests a nuanced relationship of markets and individuals' rationality: The perspective of the “No Trade Theorem”, which suggests that some irrationality is required for the smooth functioning of Prediction Markets

4 Outline

- What is the CJT
- Prediction Markets as Judgment Aggregators
- Modelling Prediction Markets as Exchange Economies
- Jury Theorems for Prediction Markets
- The Perspective of the “No Trade Theorem”

5 What is the CJT and What Are Prediction Markets

(Sadly, the fun part of the talk is over now, and we are now entering the drudgery of some very formal results)

5.1 General Setting

We have a proposition q whose truth value $\varphi(q) \in \{0, 1\}$ is (as of yet) unknown. We have a population of n agents (for simplicity, assume n is odd), whose beliefs about q we denote with $B_1(q), \dots, B_n(q)$. Let $V(q) \in \{0, 1\}$ denote the result of a majority vote taken among the agents.

5.2 Assumptions Required for the CJT

- **Competence:** Each agent has the same, better-than-chance probability $r > \frac{1}{2}$ of getting it right — $\mathbb{P}(B_i(q) = x \mid \varphi(q) = x) = r > \frac{1}{2}$.
- **Independence:** Agents' beliefs are stochastically (mutually) independent, conditional on the state of the world.
- **Sincerity:** Every agent casts their vote in accordance with what they truly believe. I.e. $V(q) = 1$ iff $\#\{i \mid B_i(q) = 1\} > \#\{i \mid B_i(q) = 0\}$.

5.3 Theorem 1 (Condorcet Jury Theorem)

For each possible truth value of q , $x \in \{0,1\}$, the probability of a correct majority vote $\mathbb{P}(V(q) = x \mid \varphi(q) = x)$

- is greater than each individual's probability of having the correct belief $\mathbb{P}(B_i(q) = x \mid \varphi(q) = x) = r$ (“non-asymptotic result”), and
- converges to 1 as the number of participating individuals increases (“asymptotic result”).

5.4 Prediction Markets

- A prediction market on q is a market trading bets on q and $\neg q$.
 - Let b_q be an asset that pays out €1 if q and €0 if $\neg q$ (and vice versa).
 - Let p_q and $p_{\neg q}$ denote the equilibrium prices of these assets.

5.5 Prediction Markets as Judgment Aggregators

- Let $M(q) = \chi(p_q > p_{\neg q})$ denote “the market’s judgment”: It affirms q iff the market price of a bet on q is higher than the market price of a bet on $\neg q$.

5.6 Example



Figure 2: (“Will 2025 Be the Hottest Year Ever?” 2025)

6 Modelling Prediction Markets as Exchange Economies

6.1 Definition 1 (Exchange Economy)

An **exchange economy** is an economy in which there are k commodities and n consumers $\{1, \dots, n\}$, where each consumer i is described by

- a utility function $U^i(x_1^i, \dots, x_k^i)$, assigning a non-negative real number to each possible consumption bundle.
- an initial endowment of commodities $\varpi^i = (\varpi_1^i, \dots, \varpi_k^i) \in \mathbb{R}_{\geq 0}^k$.

6.2 Definition 2 (Consumption Bundle)

A **consumption bundle** for consumer i is a tuple of quantities of the commodities in the economy, $x^i = (x_1^i, \dots, x_k^i) \in \mathbb{R}_{\geq 0}^k$.

6.3 Definition 3 (Allocation in an Exchange Economy)

An **allocation** $x = (x^1, \dots, x^n) \in (\mathbb{R}_{\geq 0}^k)^n$ assigns a consumption bundle to each consumer.

6.4 Definition 4 (Feasibility of an Allocation)

An allocation x in an exchange economy with initial endowments $\varpi^1, \dots, \varpi^n$ is **feasible** if

$$\forall l \in \{1, \dots, k\} : \sum_{i=1}^n x_l^i \leq \sum_{i=1}^n \varpi_l^i$$

6.5 Definition 5 (Walrasian Equilibrium)

A **Walrasian Equilibrium** in an exchange economy is a pair $(\bar{x}, \bar{p})$, consisting of an allocation $\bar{x} = (\bar{x}^1, \dots, \bar{x}^n)$ and a price vector $\bar{p} = (\bar{p}_1, \dots, \bar{p}_k)$ such that:

- $\bar{x}^i$ maximizes $U^i(x^j)$ subject to $\bar{p} \cdot \bar{x}^i \leq \bar{p} \cdot \varpi^i$
- $\bar{x}$ is feasible.

7 Jury Theorems for Prediction Markets

7.1 Assumptions Required

- **Competence**
- **Independence**
- **Sincerity**
- **Rational Self-Interest:** Agent i purchases bundle $\bar{x}$ iff $\bar{x}$ maximizes U^i among all the bundles that i can afford.

7.2 Further Modelling Assumptions

- **Utility function:** $U^i(x_q^i, x_{-q}^i) = B_i(q)x_q^i + (1 - B_i(q))x_{-q}^i$
- **Endowment:** Let every agent initially be endowed with one b_q and one b_{-q} (i.e. one euro).

8 Jury Theorems for Prediction Markets

8.1 Theorem 2 (PMJT for Agents with Binary Beliefs)

Under the above assumptions:

For each possible truth value of q , $x \in \{0, 1\}$, the probability of a correct market judgment $\mathbb{P}(M(q) = x \mid \varphi(q) = x)$

- is greater than each individual's probability of having the correct belief $\mathbb{P}(B_i(q) = x \mid \varphi(q) = x) = r$ (“non-asymptotic result”), and
- converges to 1 as the number of participating individuals increases (“asymptotic result”).

8.2 Theorem 3 (PMJT for Agents with Degrees of Belief)

If we let $B_i(q)$ take values in the interval $[0, 1]$ instead of just in the set $\{0, 1\}$, but keep all the other assumptions the same, then:

A Walrasian equilibrium exists, and the market affirms q iff the *median credence* is higher than 0.5, and $\neg q$ otherwise. That is, the market affirms the correct proposition iff the median credence is on the correct side of the 0.5 mark.

8.3 Can We Recover the Condorcet Properties in this Case?

Question: How to translate the **competence** assumption?

We recover both properties if we assume what (Wollesen forthcoming) calls “minimal fidelity of credences”: Agents' credences are drawn from a distribution on $[0, 1]$ whose *median* is on the correct side of the 0.5 mark.

8.4 Corollary 4 (Condorcet-Like Property for PM, Degrees of Belief)

If minimal fidelity of credences is satisfied:

For each possible truth value of q , $x \in \{0, 1\}$, the probability of a correct market judgment $\mathbb{P}(M(q) = x \mid \varphi(q) = x)$

- is greater than each individual agent's probability of having a credence on the correct side of 0.5, $\mathbb{P}(|B_i(q) - x| < 0.5 \mid \varphi(q) = x)$ and
- converges to 1 as the number of agents increases.

9 The Perspective of the “No Trade Theorem”

- Consider an economy like the one we have been discussing. But assume that agents are risk-averse, have common knowledge of each other’s rationality and of this fact, non-extreme credences, and only accept trades that in (their) expectation don’t decrease their utility. Then no trade takes place.

10 Conclusion

- Prediction Markets have similar positive properties to majority voting
- Unlike majority voting, they don’t have to rely on agents’ sincerity
- But they have to be paid for in the following sense: Someone needs to enter into not fully rational trades

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